Zhuoran Xu

zornxu6.626@gmail.com

Summary

I am currently an undergraduate student in Shandong University. I am interested in using different experimental measurement to study novel quantum materials especially superconductors. I fabricated 2D material by mechanical exfoliation and used some optical measurement to study 2D material in my second year. Currently, I use scanning SQUID microscopy to study superconductors. In my spare time, I like to play the guitar and play soccer with my friends.

Education

2021 - present Undergraduate Student at **Shandong University** (GPA: 4.1/5.0)
2023 - present Exchange Student at **Bar Ilan university**, Israel

Research Experience

Xiaotao Hao Lab, Shandong University Oct 2022 - Nov 2023

The exciton property in monolayer and multilayer MoS_2

- 1. Did the mechanical exfoliation and Au-assisted mechanical exfoliation to get monolayer MoS_2
- 2. Did the measurement of PL spectrum and Raman spectrum of MoS_2
- 3. Fabricated VdWs heterostructures of MoS_2 / WS_2

Danfeng Li Lab, City University of Hong Kong Jul 2023 - Aug 2023

The fabrication and characterization of nickelate superconductor

- 1. Fabricated the nickelate superconductor
- 2. Did the XRD measurement of the sample and analyse the results

Beena Kalisky Lab, Bar Ilan university Dec 2023 - present

Scanning SQUID study of superconducting network

- 1. Learned the measument of scanning SQUID
- 2. Use machine learning methods to fit SQUID kernel of susceptometry of Josephson-junctions arrays

Honors And Rewards

Outstanding Student Scholarship in Shandong University, First class 2022

China Undergraduate Mathematical Contest in Modeling, Provincial First Class 2022

Mathematics Competition in Chinese Undergraduates, Provincial First Class 2022

Merit Scholarships in Research and Innovation in Shandong University, Second Class 2023

Outstanding Student Scholarship in Shandong University, Second class 2023

Skills

Programming Languages: Python (scientific programming and simulation), MATLAB (data processing), C
Languages: Chinese, English(IELTS 7.0), French(basic), Hebrew(basic)
Experimental techniques: Cryogenic-temperature refrigerators, Scanning SQUID microscopy, XRD, Mechanical Exfoliation, Optical measurement, Lab electronics, Electric transport measurement